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MARITIME VEHICLE WITH RECESSED TOW POINTS

Abstract

A tow assembly for a maritime vehicle and a maritime vehicle including a hull, a first opening formed in the hull, and the tow assembly, the tow assembly coupled to the hull. The tow assembly includes a housing disposed in the first opening of the hull, the housing defining a cavity recessed relative to an exterior surface of the hull. The tow assembly also includes a bar coupled to the housing at a position within the cavity, such that the bar is recessed relative to the exterior surface of the hull. The bar defines one or more tow points for the maritime vehicle.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION [0001] The present application claims priority to U.S. Provisional Patent Application No. 63/561,102, titled “Maritime Vehicle with Recessed Tow Points” and filed on Mar. 4, 2024, the contents of which are hereby incorporated by reference in its entirety.

FIELD OF THE DISCLOSURE

[0002] The present disclosure generally relates to maritime vehicles and more specifically to a maritime vehicle having recessed tow points.

BACKGROUND OF THE DISCLOSURE

[0003] Maritime vehicles, or vehicles designed for use on or in the water, are commonly used for transportation, recreation, defense, scientific research, and other purposes. Examples of maritime vehicles include boats, watercraft, submarines, and amphibious vehicles. Maritime vehicles can be manned (i.e., operated by an onboard human) or unmanned, and unmanned maritime vehicles can be remotely controlled or can be autonomous.

[0004] Boats often include one or more tow points designed to help launch, recover, or handle the boat. More particularly, tow points facilitate the pushing, pulling, or towing of the boat by a user, a towing vehicle (e.g., an automobile), or a towing apparatus through loading and unloading, no wake zones, and emergency situations. Tow points serve as the primary connection interface between the towing vehicle/apparatus and the boat being towed. Tow points also provide a reliable anchoring point, ensuring stability and control during towing maneuvers. Additionally, tow points are engineered to withstand significant loads and forces exerted during pushing/pulling/towing maneuvers, thereby preventing structural damage to the boat. Further, tow points can be used to push, pull, or tow other vehicles (e.g., other boats), people (e.g., water skiers), or other objects at a desired location (e.g., behind) the boat.

[0005] Known tow points generally take the form of a sturdy metal fixture that is attached to an exterior surface of the hull or deck of the boat and includes one or more eyelets, lops, or hooks that protrude outward from the hull or deck and can be securely connected to tow ropes, cables, or harnesses. FIG. 1A illustrates one example of a known tow point, in the form of a metal hook **52** mounted to an exterior surface of a known boat **50**. FIG. 1B illustrates another example of a known tow point, in the form of a metal plate **60** that can be mounted to the exterior surface of the known boat **50** (or a different boat) and includes a curved ring **62** that protrudes outward and is engageable by a tow rope, a cable, a harness, or a user of the boat. And FIG. 1C illustrates another example of a known tow point, in the form of a tow bar **70** that can be attached to an exterior surface of a boat and protrudes outward so as to be engageable by a tow rope, a cable, a harness, or a user of the boat.

SUMMARY

[0006] A tow assembly for a maritime vehicle and a maritime vehicle including a hull, a first opening formed in the hull, and the tow assembly, the tow assembly coupled to the hull. The tow assembly includes a housing disposed in the first opening of the hull, the housing defining a cavity recessed relative to an exterior surface of the hull. The tow assembly also includes a bar coupled to the housing at a position within the cavity, such that the bar is recessed relative to the exterior surface of the hull. The bar defines one or more tow points for the maritime vehicle.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0007] The features of this invention which are believed to be novel are set forth with particularity

in the appended claims. The invention may be best understood by reference to the following description taken in conjunction with the accompanying drawings, in which like reference numerals identify like elements in the several FIGS., in which:

[0008] FIG. 1A is a perspective view of a portion of one example of a known maritime vehicle having one example of a known tow point;

[0009] FIG. 1B is a perspective view of another example of a known tow point;

[0010] FIG. 1C is a perspective view of another example of a known tow point;

[0011] FIG. 2A is a top perspective view of an example of a maritime vehicle constructed in accordance with the teachings of the present disclosure;

[0012] FIG. 2B is a front view of the maritime vehicle of FIG. 2A;

[0013] FIG. 2C is a rear view of the maritime vehicle of FIG. 2A;

[0014] FIG. 2D is a bottom perspective view of the maritime vehicle of FIG. 2A;

[0015] FIG. 2E is another bottom perspective view of the maritime vehicle of FIG. 2A;

[0016] FIG. 2F is similar to FIG. 2A, but with the cap and various components of the maritime vehicle removed for illustrative purposes;

[0017] FIG. 2G is similar to FIG. 2F, but with additional components of the maritime vehicle removed for illustrative purposes;

[0018] FIG. 2H is a rear, perspective view of the maritime vehicle of FIG. 2F;

[0019] FIG. 2I is a first cross-sectional view taken along line I-I in FIG. 2A;

[0020] FIG. 2J is a second cross-sectional view taken along line J-J in FIG. 2A;

[0021] FIG. 2K is similar to FIG. 2A, but with the latching assembly of the maritime vehicle partially removed for illustrative purposes;

[0022] FIG. 3A is similar to FIG. 2C, but shows how the maritime vehicle of FIGS. 2A-2K can include tow assemblies constructed in accordance with the teachings of the present disclosure;

[0023] FIG. 3B is a close-up, perspective view of one of the tow assemblies shown in FIG. 3A;

[0024] FIG. 3C is a rear view of the tow assembly of FIG. 3B removed from the maritime vehicle;

[0025] FIG. 3D is a front view of the tow assembly of FIG. 3C, showing a plate of the tow assembly;

[0026] FIG. 3E is similar to FIG. 3D, but shows the tow assembly with a different plate coupled thereto;

[0027] FIG. 3F is similar to FIG. 3B, but shows an antenna holder coupled to the tow assembly; and

[0028] FIG. 3G is similar to FIG. 3C but shows the tow assembly with a different plate coupled thereto.

DETAILED DESCRIPTION

[0029] The present disclosure is directed to a maritime vehicle that is primarily intended for use for military purposes (e.g., for naval defense, patrolling waters and enforcing laws, reconnaissance, naval exploration, monitoring) but can also be used for other purposes if desired. The maritime vehicle is small(er), durable, and portable, both in use and when not in use but in transit between two different locations (e.g., two bodies of water). In use, the maritime vehicle is configured to quickly, efficiently, and stealthily traverse a body of water once dispatched (e.g., from other maritime vehicles, beachheads, or an airdrop). The maritime vehicle is modular, with components that can be flexibly altered, removed, or added as desired in accordance with the mission of the maritime vehicle. The maritime vehicle can collaborate with other similar maritime vehicles and/or military assets when necessary. The maritime vehicle is preferably unmanned and autonomous though need not be.

[0030] FIGS. 2A-2K and 3A-3G illustrate one example of a maritime vehicle **100** constructed in accordance with the teachings of the present disclosure. The maritime vehicle **100** is an unmanned vessel configured to autonomously traverse a body of water. The maritime vehicle **100** generally includes a hull **104** and a cap **108** that is coupled to the hull **104** to secure various components

within the maritime vehicle **100**. The hull **104** is at least partially disposed in the body of water in which the maritime vehicle **100** is traversing. The hull **104** in this example is a mono-hull that has a front (or bow) **112**, a rear (or stern) **116**, two sides **120**, and a keel **124** coupled to another. The front **112**, the rear **116**, the sides **120**, and the keel **124** can be welded together or can be coupled to one another in a different manner. For example, the front **112**, the rear **116**, the sides **120**, and the keel **124** can be coupled together in the manner described in U.S. Provisional Application No.

63/561,282, titled "Systems and Approaches for Assembling a Maritime Vehicle" and filed Mar. 4, 2024, the contents of which are hereby incorporated by reference herein. The hull **104** is configured such that the hull provides a continuous planning surface that allows the maritime vehicle **100** to be highly maneuverable and to ride along the top of a body of water at high speeds, even in extreme weather conditions and difficult to navigate bodies of water. Meanwhile, the cap **108** is coupled to the hull **104** to cover and/or conceal the components of the maritime vehicle **100** disposed in and carried by the hull **104** as the maritime vehicle **100** traverses the body of water.

[0031] In this example, the hull **104** and the cap **108** each have a length that is equal to approximately 6 feet. In other examples, however, the length can vary. For example, the length can be equal to approximately 14 feet. The hull **104** is preferably entirely made of aluminum but can be partially or entirely be made of fiberglass and/or one or more other materials. In other examples, the maritime vehicle **100** can include two or more hulls (e.g., two parallel hulls) instead of the mono-hull. In this example, the cap **108** entirely covers the hull **104** (and the components therein). In other examples, however, the maritime vehicle **100** need not include the cap **108** or the cap **108** may only partially cover the hull **104** (and the components disposed therein).

[0032] In some examples, the cap **108** can be removably coupled to the hull **104** via a locking system. For example, as illustrated in FIGS. 2A-2K, the locking system can take the form of a plurality of latch mechanisms **128** disposed around a perimeter of the maritime vehicle **100**. Thus, the cap **108** can be removed to allow access to the interior of the hull **104**. In other examples, however, the cap **108** can be permanently coupled to the hull **104** to permanently conceal the components within the maritime vehicle **100**.

[0033] The maritime vehicle **100** also includes a plurality of bulkheads **132** arranged within the hull **104**. The bulkheads **132** divide the maritime vehicle **100** into a plurality of different compartments for receiving and retaining different components in the maritime vehicle **100**.

[0034] The maritime vehicle **100** also includes a sensor system that is generally configured to collect data about various components of the maritime vehicle **100** as well as data about the environment surrounding the maritime vehicle **100** (including data about objects in that environment). To this end, the sensor system generally includes a plurality of sensors disposed on an exterior or an interior of the maritime vehicle **100**. The sensors can include, for example, one or more pressure sensors (e.g., positioned to detect the pressure of the ambient air external to the maritime vehicle **100**, the pressure of the water in which the maritime vehicle **100** is disposed, the pressure within the maritime vehicle **100**), one or more temperature sensors (e.g., positioned to measure a temperature of a component of the maritime vehicle **100**, a temperature of ambient air external to the maritime vehicle **100**, a temperature of water in which the maritime vehicle **100** is disposed), one or more acoustic sensors (e.g., sonar sensors), one or more LIDAR sensors, one or more location sensors (e.g., GPS sensors, compass sensors), one or more motion sensors (e.g., accelerometers, gyroscopes), one or more infrared sensors, one or more water sensors (e.g., a float switch, a capacitive sensor, an ultrasonic sensor, an electrical water sensor, etc.) to determine when water is present and/or present to a given extent (e.g., at a certain volume or level), one or more humidity sensors, one or more power sensors (e.g., configured to detect charging or fueling levels), one or more lighting sensors (e.g., daylight sensors), one or more imaging sensors (e.g., CCD sensors, CMOS sensors), one or more magnetic sensors, or combinations thereof.

[0035] The maritime vehicle **100** also includes a power system that is generally configured to power the maritime vehicle **100** (and the components of the maritime vehicle **100**). The power

system generally includes a thrust system and one or more power sources configured to power the thrust system (and the other components within the maritime vehicle **100**). The thrust system is generally configured to propel the maritime vehicle **100** in/on/along the water. The thrust system can be a propeller-based thrust system or can be a jet pump-based thrust system such as the jet pump assembly described in U.S. Provisional Application No. 63/561,166, titled “Jet Pump Assembly for Maritime Vehicle” and filed Mar. 4, 2024, the contents of which are hereby incorporated by reference herein. The one or more power sources can include, for example, one or more batteries, fuel (e.g., gasoline, diesel) stored in tanks carried by the maritime vehicle **100**, hydrogen stored in hydrogen tanks carried by the maritime vehicle **100**, solar panels (e.g., mounted to an exterior of the vehicle **100**), or other sources. The maritime vehicle **100** illustrated in FIGS. 2A-2K includes four battery assemblies each including a rechargeable battery. The maritime vehicle **100** illustrated in FIGS. 2A-2K also includes a retention assembly for the four battery assemblies, e.g., the retention assembly described in U.S. Provisional Application No. 63/561,063, titled “Power System for Maritime Vehicle” and filed Mar. 4, 2024, the contents of which are hereby incorporated by reference herein. The maritime vehicle **100** generally also includes a cooling system configured to cool the thrust system and/or the one or more power sources, thereby preventing these components from overheating and leading to failure of the maritime vehicle **100**. For example, the maritime vehicle **100** can include the cooling system described in U.S. Provisional Application No. 63/561,181, titled “Micro-Keel Cooler for Maritime Vehicle” and filed Mar. 4, 2024, the contents of which are hereby incorporated by reference herein.

[0036] In operation, the maritime vehicle **100** may be used to deploy and/or retrieve payloads such as, for example, persons, weapons (e.g., drones, missiles, mines, bombs), cargo (e.g., food), scientific instruments, or other equipment. Payloads can be deployed aerially (into the air), underwater, or on the surface of the water. Payloads can also be retrieved from the air, from underwater, or the surface of the water. Payloads to be deployed can be disposed in the hull **104**, attached to the exterior surface of the hull **104**, or attached to the exterior surface of the cap **108** prior to deployment. Likewise, retrieved payloads can be stored in the hull **104**, attached to and stored on the exterior surface of the hull **104**, or attached to and stored on the exterior surface of the cap **108**.

[0037] The maritime vehicle **100** can also include other systems to help with the operation of the maritime vehicle **100**, for example a ballast system, a navigation system, and a vision system. The ballast system is generally configured to stabilize the maritime vehicle **100** in the water, regardless of whether the maritime vehicle **100** is stationary or on the move. To this end, the maritime vehicle **100** may include one or more ballast tanks or chambers selectively filled with water or air to vary the buoyancy of the maritime vehicle **100**. Alternatively or additionally, the ballast system may include and utilize one or more inflatable devices to vary the buoyancy of the maritime vehicle **100**. The ballast system may also provide for the selective submerging and re-surfacing of the maritime vehicle **100** in a similar manner. The navigation system, which may for example be an inertial navigation system, utilizes the sensors of the sensor system to track the position and orientation of the maritime vehicle **100** and to guide the maritime vehicle **100** to its desired location in the body of water (or in a different body of water). The vision system is generally configured to capture, process, and analyze images obtained by the one or more image sensors and other data (e.g., data obtained by other sensors in the sensor system). The vision system can in turn identify or classify the environment surrounding the maritime vehicle **100** (including objects in that environment).

[0038] The maritime vehicle **100** further includes a communications system that is generally configured to facilitate communication (i) between the maritime vehicle **100** and one or more central (remote) controllers, (ii) between the maritime vehicle **100** and and/or one or more other maritime vehicles **100** and/or other military assets (e.g., planes, ships), and (iii) between different components of the maritime vehicle **100**. The communications system generally includes one or

more local controllers and one or more communication modules (e.g., one or more antennae, one or more receivers, one or more transmitters, one or more radios, one or more ethernet switches) to effectuate wired or wireless communication between the maritime vehicle **100** and the central controller(s) or other maritime vehicles **100**. For example, the maritime vehicle **100** includes a plurality of antennae disposed on an exterior of the cap **108** as well as a plurality of antennae disposed in the hull **104**.

[0039] The one or more local controllers are generally configured to communicate data (e.g., operational instructions, data from the sensor system, data from other maritime vehicles **100** or military assets) and to perform automated operations of the maritime vehicle **100** based on that data. In some examples, the maritime vehicle **100** includes a plurality of different local controllers. For example, the maritime vehicle **100** can include one or more thrust controllers (for controlling the operation of the thrust system), one or more sensor controllers (for controlling the sensors in the sensor system), one or more payload controllers (for deploying or retrieving payloads), one or more navigation controllers (as part of the navigation system), and one or more ballast controllers (for controlling the ballast system). It will be appreciated that each of the one or more controllers may be implemented as hardware (e.g., processor, die, integrated device), software (e.g., non-transitory processor readable medium), and/or combinations thereof, in one or more devices (e.g., processor, chip, computer, tablet, mobile device).

[0040] While not explicitly described or illustrated herein, it will be appreciated that the maritime vehicle **100** includes several additional components. For example, the maritime vehicle **100** includes various sealing elements configured to provide seals between different components of the vehicle **100** (or between the vehicle **100** and the environment surrounding the vehicle **100**). As another example, the maritime vehicle **100** also includes various fasteners that help to couple the components of the maritime vehicle **100** together. As yet another example, the maritime vehicle **100** includes cabling that helps to communicatively couple components of the maritime vehicle **100** together. As yet another example, the maritime vehicle **100** includes various electrical components that help to operate the maritime vehicle **100**, e.g., one or more relay boards, one or more DC-DC converters, one or more supervisor boards, one or more brain boards.

[0041] Like known boats such as the known boat **50** illustrated in FIG. **1A**, the maritime vehicle **100** further includes one or more tow points configured to help launch, recover, or handle the maritime vehicle **100**. However, the tow points of the maritime vehicle **100** are different from the tow point of the known boat **50** (i.e., the metal hook **52**), the tow points illustrated in FIGS. **1B** and **1C**, and other known tow points. Instead, as best illustrated in FIGS. **2F-2H** and **3A-3D**, the tow points of the maritime vehicle **100**, which are defined by one or more tow assemblies **200**, are recessed relative to an exterior surface of the hull **104**. In other words, the tow portions of the maritime vehicle **100** are through hull tow points. Recessing the one or more tow assemblies **200** within the hull **104** significantly reduces the likelihood that the one or more tow assemblies **200** will be damaged (e.g., by a towing apparatus) and reduces the likelihood that the one or more tow assemblies **200** will cause damage (e.g., to a user). Moreover, each of the one or more tow assemblies **200** includes a plate **204** that is at least partially positioned within the interior **110** of the hull **104** and is sized to receive and retain one or more other components of the maritime vehicle **100**, such as, for example, a board (e.g., a relay board, a supervisor board, a brain board) and/or an antenna holder configured to hold an antenna of the plurality of antennae.

[0042] In the example illustrated in FIGS. **2A-2K** and **3A-3D**, the maritime vehicle **100** includes two tow assemblies **200**—a first tow assembly **200A** and a second tow assembly **200B**. The first tow assembly **200A** and the second tow assembly **200B** are structurally and functionally identical to one another, with the exception that the first and second tow assemblies **200A**, **200B** are coupled to different portions of the hull **104**. In this example, the first tow assembly **200A** is coupled to a first portion of the hull **104** via a first opening **208** formed in the hull **104** and the second tow assembly **200B** is coupled to a second portion of the hull **104** via a second opening **212** formed in the hull

104. In particular, the first opening **208** is partially formed in the stern **116** of the hull **104** and partially formed in one of the sides **120** of the hull **104**, such that the first tow assembly **200A** is located at a first, rear corner of the hull **104**, whereas the second opening **212** is partially formed in the stern **116** of the hull **104** and partially formed in the other of the sides **120** of the hull **104**, such that the second tow assembly **200B** is located at a second, rear corner of the hull **104** opposite the first, rear corner of the hull **104**. So positioned, the first and second tow assemblies **200A** and **200B** not only provide reliable anchoring points, ensuring stability and control during towing maneuvers, but the tow points defined by the first and second tow assemblies **200A** and **200B** are able to withstand greater loads and forces than known tow points. Additionally, it will be appreciated that the two tow assemblies **200** can be used for different purposes. For example, the first tow assembly **200A** can be used to tie the maritime vehicle **100** to another object while the second tow assembly **200B** can be used to pick-up the maritime vehicle **100**.

[0043] As best illustrated in FIGS. 3C and 3D, the first tow assembly **200A** includes a housing **216** and a bar **220** that is coupled to the housing **216** and specifically defines tow points for pushing, pulling, or towing the maritime vehicle **100** or pushing, pulling, or towing other vehicles (e.g., other maritime vehicles), people, or other objects. The housing **216** is generally sized and shaped to be disposed in the first opening **208** formed in the hull **104**. More particularly, the housing **216** has an irregular shape defined by a top wall **224**, a bottom wall **228**, a rear wall **232**, and two side walls **236**, **240**. As best illustrated in FIG. 3C, the top wall **224** and the bottom wall **228** each have a substantially triangular shape, whereas the rear wall **232** and the side walls **236**, **240** each have a rectangular shape. The housing **216** also includes a recessed cavity **244** defined by the walls **224**, **228**, **232**, **236**, **240**. The recessed cavity **244** is recessed relative to an exterior surface **246** of the hull **104**. The recessed cavity **244** is also recessed relative to outwardly facing perimeter edges of the walls **224**, **228**, **236**, **240**. Meanwhile, the bar **220** is coupled to the housing **216** and is sized to allow a user of the maritime vehicle **100** to grip the bar **220** and move (e.g., pick-up, pull) the maritime vehicle **100**. The bar **220** is also sized and arranged to serve as a reliable and strong anchoring point for a vehicle (e.g., an automobile, another maritime vehicle), a towing apparatus, people (e.g., water skiers), or other objects. More particularly, the bar **220** is welded to the walls **224-240** at a position within the recessed cavity **244**. In this example, the bar **220** has three portions—a first bar portion **248**, a second bar portion **252**, and a third bar portion **256**. The first bar portion **248** is directly coupled to and extends between the top and bottom walls **224**, **228** at a corner of the hull **104**. As such, the first bar portion **248** extends in a vertical direction. The second bar portion **252** is directly coupled to and extends between the side wall **236** and the first bar portion **248**, while the third bar portion **256** is directly coupled to and extends between the side wall **240** and the first bar portion **248**. As such, each of the second and third bar portions **252**, **256** extends in a horizontal direction that is generally perpendicular to the first bar portion **248**.

[0044] As best illustrated in FIG. 3D, the first tow assembly **200A** also includes the plate **204**, which can, for example, be used to mount other components of the maritime vehicle **100** to the first tow assembly **200A**. The plate **204** is coupled to an exterior surface of the rear wall **232** such that the plate **204** is positioned outside of the recessed cavity **244**. In this example, the plate **204** is a solid plate having a rectangular shape. In other examples, however, the plate **204** can have a different shape and/or include one or more mounting apertures for mounting components to the plate **204**. For example, as illustrated in FIGS. 3E and 3F, the plate **204** can include four mounting apertures **264** sized to respectively receive four fasteners (not shown) to mount an antenna holder **268** to the plate **204** (and, thus, to the first tow assembly **200A** and the maritime vehicle **100** more generally). Likewise, the size of the plate **204** can vary. For example, as illustrated in FIGS. 3E and 3F, the plate **204** can be shorter in height than the plate **204** illustrated in FIGS. 3C and 3D. As another example, the plate **204** can be shorter than height than the plate **204** illustrated in FIGS. 3C-3F such that the plate **204** is no longer visible from behind the first tow assembly **200A**).

[0045] As briefly discussed above, and as best illustrated in FIGS. 2F-2H, 3A, and 3B, the housing

216 of the first tow assembly **200A** is disposed in the first opening **208** in the hull **104**. When the housing **216** of the first tow assembly **200A** is disposed in the first opening **208**, the side wall **236** of the housing **216** engages a portion of the stern **116** of the hull **104** and the side wall **238** of the housing **216** engages a portion of one of the side walls **120** of the hull **104**. Meanwhile, the top wall **224** of the housing **216** engages a portion of the hull **104** and extends over an interior portion of the hull **104**, whereas the bottom wall **228** of the housing **216** engages a portion of the stern **116** of the hull **104** and a portion of one of the side walls **120** of the hull **104**. In turn, the recessed cavity **244** of the first tow assembly **200A** faces rearward and outward (i.e., toward the environment surrounding the maritime vehicle **100**), whereas the plate **204** of the first tow assembly **200A** is disposed within and faces the interior **110** of the hull **104**. As such, the plate **204** is optimally positioned to receive and hold a relay board **272** (see FIG. 2F), a supervisor board, or another electronic component of the maritime vehicle **100**.

[0046] As also briefly discussed above, the second tow assembly **200B** is structurally and functionally identical to the first tow assembly **200A**. Thus, further details about the second tow assembly **200B** are omitted for brevity. It will also be appreciated that in other examples, the maritime vehicle **100** can include only one tow assembly **200** or can include more than two tow assemblies **200**. It will also be appreciated that in other examples, the tow assemblies **200** can be coupled to different portions of the hull **104**. For example, a tow assembly **200** can be coupled to the hull **104** via an opening partially or entirely formed in the bow of the hull **104**.

[0047] Finally, although certain maritime vehicles have been described herein in accordance with the teachings of the present disclosure, the scope of coverage of this patent is not limited thereto. On the contrary, while the invention has been shown and described in connection with various preferred embodiments, it is apparent that certain changes and modifications, in addition to those mentioned above, may be made. This patent covers all embodiments of the teachings of the disclosure that fairly fall within the scope of permissible equivalents. Accordingly, it is the intention to protect all variations and modifications that may occur to one of ordinary skill in the art.

Claims

1. A maritime vehicle, comprising: a hull; a first opening formed in the hull; and a tow assembly coupled to the hull, the tow assembly comprising: a housing disposed in the first opening of the hull, the housing defining a cavity recessed relative to an exterior surface of the hull; and a bar coupled to the housing at a position within the cavity, such that the bar is recessed relative to the exterior surface of the hull, wherein the bar defines one or more tow points for the maritime vehicle.
2. The maritime vehicle of claim 1, wherein the hull is a mono-hull.
3. The maritime vehicle of claim 1, wherein the first opening is formed in a stern of the hull.
4. The maritime vehicle of claim 1, wherein the hull comprises a bow, a stern, two sides, and a keel coupled to one another, and wherein the first opening is partially formed in the stern and partially formed in one of the two sides.
5. The maritime vehicle of claim 1, the tow assembly further comprising a plate coupled to the housing at a position outside of the cavity.
6. The maritime vehicle of claim 5, further comprising a relay or supervisor board coupled to the plate.
7. The maritime vehicle of claim 5, wherein the plate is disposed within an interior of the hull.
8. The maritime vehicle of claim 5, the tow assembly further comprising an antenna holder sized to hold an antenna therein, the antenna holder coupled to the plate.
9. The maritime vehicle of claim 1, wherein the bar is welded to the housing.
10. The maritime vehicle of claim 1, wherein the bar comprises a first bar portion and a second bar

portion, wherein the first bar portion extends in a first direction and the second bar portion extends in a second direction substantially perpendicular to the first direction.

11. The maritime vehicle of claim 1, the tow assembly further comprising an antenna holder sized to hold an antenna therein.

12. A maritime vehicle, comprising: a hull; a first opening formed in a first portion of the hull; a second opening formed in a second portion of the hull; a first tow assembly coupled to the first portion of the hull, the first tow assembly comprising: a first housing disposed in the first opening of the hull, the housing defining a first cavity recessed relative to an exterior surface of the hull; and a first bar coupled to the first housing at a position within the first cavity, such that the first bar is recessed relative to the exterior surface of the hull, wherein the first bar defines one or more first tow points for the maritime vehicle; and a second tow assembly coupled to the second portion of the hull, the second tow assembly comprising: a second housing disposed in the second opening of the hull, the second housing defining a second cavity recessed relative to the exterior surface of the hull; and a second bar coupled to the second housing at a position within the second cavity, such that the second bar is recessed relative to the exterior surface of the hull, wherein the second bar defines one or more second tow points for the maritime vehicle.

13. The maritime vehicle of claim 12, wherein each of the first and second openings is formed in a stern of the hull.

14. The maritime vehicle of claim 12, wherein the hull comprises a bow, a stern, two sides, and a keel coupled to one another, wherein the first opening is partially formed in the stern and partially formed in one of the two sides, and wherein the second opening is partially formed in the stern and partially formed in the other of the two sides.

15. The maritime vehicle of claim 12, the first tow assembly further comprising a first plate coupled to the first housing at a position outside of the first cavity.

16. The maritime vehicle of claim 15, wherein the first plate is disposed within an interior of the hull.

17. The maritime vehicle of claim 15, the first tow assembly further comprising an antenna holder sized to hold an antenna therein, the antenna holder coupled to the first plate.

18. The maritime vehicle of claim 12, wherein the first bar is welded to the first housing.

19. The maritime vehicle of claim 12, wherein the first bar comprises a first bar portion and a second bar portion, wherein the first bar portion extends in a first direction and the second bar portion extends in a second direction substantially perpendicular to the first direction.

20. A tow assembly for a maritime vehicle, the tow assembly comprising: a housing adapted to be disposed in a first opening of a hull of the maritime vehicle, the housing defining a cavity recessed relative to an exterior surface of the hull; and a bar coupled to the housing at a position within the cavity, such that the bar is recessed relative to the exterior surface of the hull, wherein the bar defines one or more tow points for the maritime vehicle.
